

Quantum Bayesian Networks: Compositionality and Typing via Linear Logic

Rémi Di Guardia, Thomas Ehrhard, Claudia Faggian

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LIQCS, 18 June 2026



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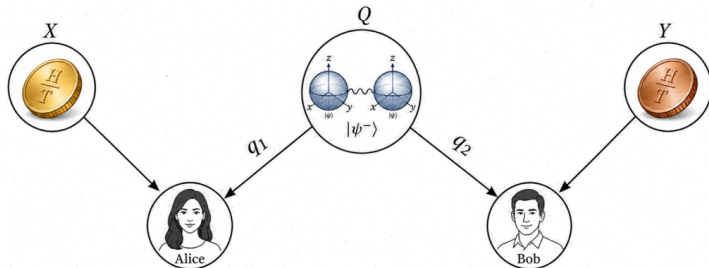
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Quantum – Bell's Experiment



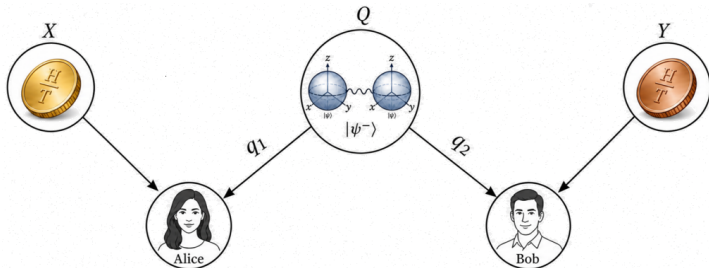
Q = Preparation of the entangled q_1 and q_2

X = Flip a coin

Alice = Measure on q_1 parameterized by the value of X

Question: What is $\Pr(\text{Alice} = 0, \text{Bob} = 0 \mid X = 1, Y = 0)$?

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General Problem: Considering a quantum system, what is the probability of obtaining a given measurement?

Bayesian Networks

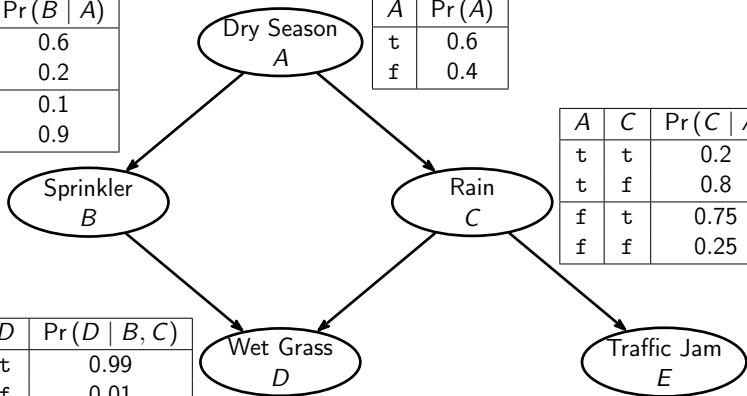
A	B	$\Pr(B A)$
t	t	0.6
t	f	0.2
f	t	0.1
f	f	0.9

A	$\Pr(A)$
t	0.6
f	0.4

A	C	$\Pr(C A)$
t	t	0.2
t	f	0.8
f	t	0.75
f	f	0.25

B	C	D	$\Pr(D B, C)$
t	t	t	0.99
t	t	f	0.01
t	f	t	0.7
t	f	f	0.3
f	t	t	0.9
f	t	f	0.1
f	f	t	0.01
f	f	f	0.99

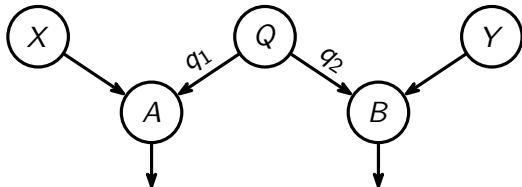
C	E	$\Pr(E C)$
t	t	0.7
t	f	0.3
f	t	0.1
f	f	0.9



Quantum Bayesian Networks

Definition: Quantum Bayesian Networks [Henson, Lal, Pusey 2014]

DAG with a family of **quantum operations** per node such that composing these top-down yields a *probability distribution*.



$Q : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$

= Preparation of the entangled q_1 and q_2

$X : \text{a map } f_x : \mathbb{C} \rightarrow \mathbb{C} \text{ for each value } x \text{ of } X$
(with $\sum_x f_x(1) = 1$)

= Flip a coin

$A : \text{a map } g_{x,a} : \mathcal{H}_1 \rightarrow \mathbb{C} \text{ for each value } x \text{ of } X \text{ and } a \text{ of } A$
(with $\sum_a g_{x,a}(q_1) = 1$)

= Measure on q_1 parameterized by the value of X

Uses

Motivations: study probabilities of systems both quantum & classical

- **Conditional Independence / No-Signaling**

When are *classical* data independent? conditionally independent?

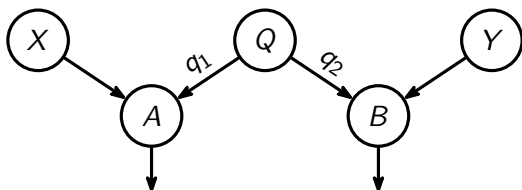
→ *d-separation*, a purely graphical criterion

Example: A and Y independent in Bell's experiment, but A and Y not independent given B

- **In which situations is quantum stronger than classical?**

Well-known that in Bell's Experiment quantum $>$ classical

→ DAGs where quantum $>$ classical? where quantum = classical?



Limits

Problem: lack of two properties

- **Compositionality**

Given a *decomposition* of a network, is the data of the full network obtained from the data of each *part*?

→ a big advantage of a *graphical* syntax, and a main property of* Bayesian networks

- **Modularity**

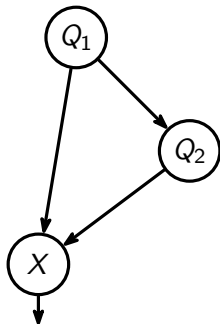
Given two *parts*, do they *compose* into a network?

→ the result must be a *DAG*

Our contributions:

Solutions for these two problems, by giving another presentation of Quantum Bayesian Networks

Limits – Compositionality



$$\phi^{Q_1} : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

$$\phi^{Q_2} : \mathcal{H}_2 \rightarrow \mathcal{H}_3$$

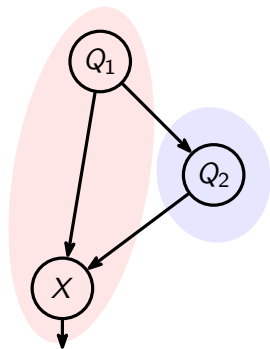
$$\phi^X(x) : \mathcal{H}_1 \otimes \mathcal{H}_3 \rightarrow \mathbb{C}$$

Full network:

$$\phi^{Q_1, Q_2, X}(x) : \mathbb{C} \rightarrow \mathbb{C}$$

$$= \phi^X(x) \circ (id_{\mathcal{H}_1} \otimes \phi^{Q_2}) \circ \phi^{Q_1}$$

Limits – Compositionality



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$$\phi^{Q_1, X}(x) : \mathcal{H}_3 \rightarrow \mathcal{H}_2$$

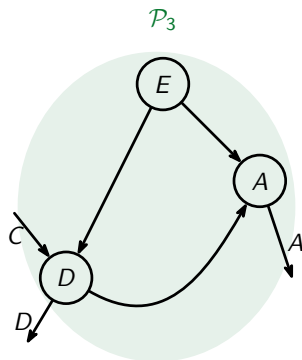
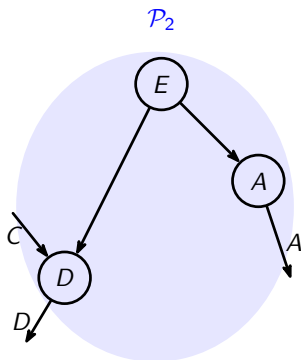
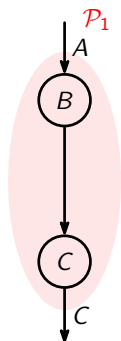
$$\phi^{Q_2} : \mathcal{H}_2 \rightarrow \mathcal{H}_3$$

$\Rightarrow \phi^{Q_1, Q_2, X}(x)$ is not the **composition** of $\phi^{Q_1, X}(x)$ and ϕ^{Q_2} !

(typed either $\mathcal{H}_3 \rightarrow \mathcal{H}_3$ or $\mathcal{H}_2 \rightarrow \mathcal{H}_2$)

Idea: Functions do not work well at graph level, **matrices** would be better (also in Bayesian networks: factors instead of conditional probabilities)

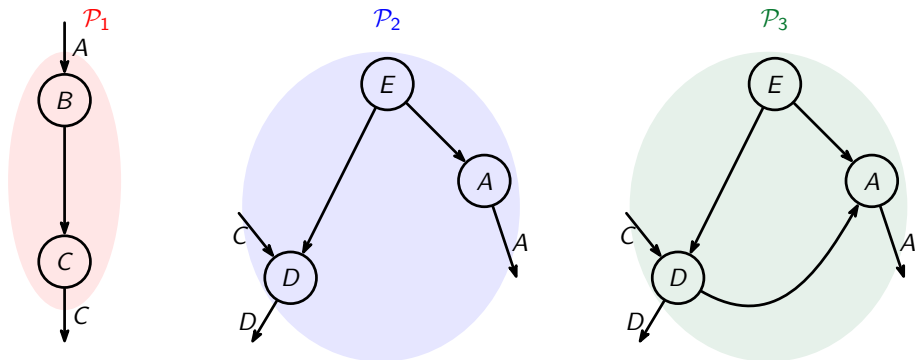
Limits – Modularity



- Part P_1 waits for input A and outputs C
- Parts P_2 and P_3 wait for input C and output A and D

Is it “legal” to branch P_1 to P_2 ? P_1 to P_3 ?

Limits – Modularity



- Part P_1 waits for input A and outputs C
- Parts P_2 and P_3 wait for input C and output A and D

Is it “legal” to branch P_1 to P_2 ? P_1 to P_3 ?

→ $P_1 \cup P_2$ is a QBN but $P_1 \cup P_3$ is not (cycle $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$)

Idea: Inputs & Outputs are insufficient, we need a **type**
Instead of a new syntax use **proof-nets** = typed graphs from Linear Logic

Plan

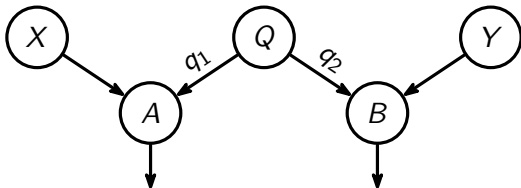
- ▶ Compositionality by Quantum Factors
- ▶ Modularity by Typing via Linear Logic

Quantum Factors

Instead of associating *quantum operations* to a node, we associate a:

Definition: Quantum Factor

Let $(X_1, \dots, X_n)$ be random variables and $(\mathcal{H}_1, \dots, \mathcal{H}_m)$ Hilbert spaces. A **Quantum Factor** on $(X_1, \dots, X_n, \mathcal{H}_1, \dots, \mathcal{H}_m)$ is a function ϕ from $\prod_{i=1}^n \text{Val}(X_i)$ to positive matrices in $\bigotimes_{j=1}^m \mathcal{H}_j$.



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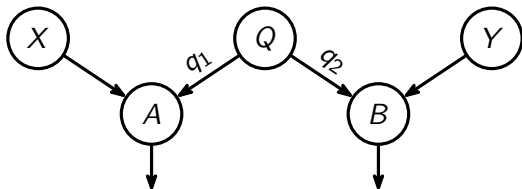
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Equipped with a **product** $\odot$ and a **sum** $\sum$

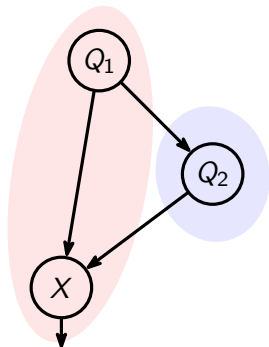
If ϕ_1 is on $(\mathbb{X}_1, \mathbb{H}_1)$ and ϕ_2 on $(\mathbb{X}_2, \mathbb{H}_2)$, $\phi_1 \odot \phi_2$ is on $(\mathbb{X}_1 \cup \mathbb{X}_2, \mathbb{H}_1 \Delta \mathbb{H}_2)$

Only quantum: get $\otimes$ -networks with $\odot$ = their product, $\sum$ = partial trace

Only classical: get factors of Bayesian networks with $\odot$ = their product, $\sum$ = their sum



Compositionality by Quantum Factors



$$\phi^{Q_1} : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

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Full network:

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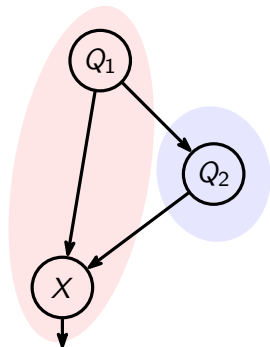
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Compositionality by Quantum Factors



$$\phi^{Q_1} : \mathcal{H}_1 \otimes \mathcal{H}_2$$

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$$\phi^X : \text{Val}(X) \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_3$$

Full network:

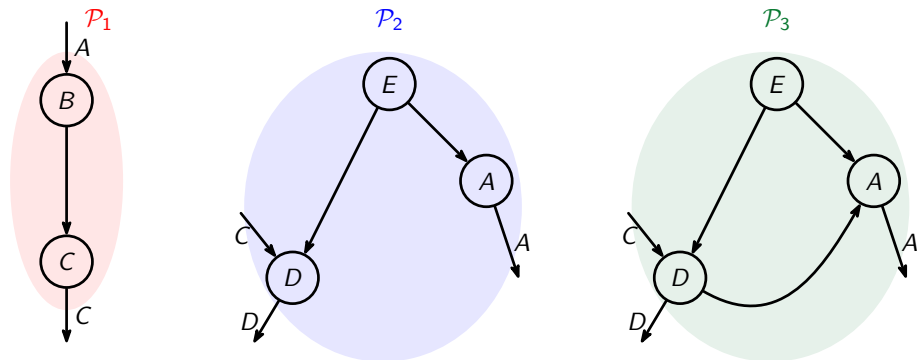
$$\begin{aligned}\phi^{Q_1, Q_2, X} : \text{Val}(X) &\rightarrow \mathbb{C} \\ &= \phi^X \odot \phi^{Q_2} \odot \phi^{Q_1}\end{aligned}$$

$$\begin{aligned}\phi^{Q_1, X} : \text{Val}(X) &\rightarrow \mathcal{H}_3 \otimes \mathcal{H}_2 \\ \implies \phi^{Q_1, Q_2, X} &= \phi^{Q_1, X} \odot \phi^{Q_2}\end{aligned}$$

$$\phi^{Q_2} : \mathcal{H}_2 \otimes \mathcal{H}_3$$

More generally, can compute for **any order** of the nodes!

Modularity



Observation: $P_1 \cup P_2$ is a QBN but not $P_1 \cup P_3$

How to ensure two parts always form a QBN?

Idea: We add a **type** = an interface

Could do so by patching QBN and getting yet another new syntax...

We prefer to use **proof-nets** = typed graphs from Linear Logic

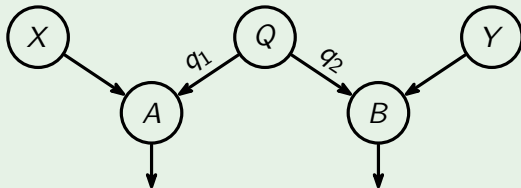
Proof-Nets for Quantum Bayesian Networks

Definition: Proof-Net

Graph respecting some **graphical criterion** and built from:

$$X^- \vdash^{ax} X^+ \quad q^- \vdash^{ax} q^+ \quad A^\perp \vdash_{cut} A \quad A \vdash_{\otimes} B \quad A \vdash_{\multimap} B \quad X^- \vdash_c X^- \quad w \quad X^- / q^- \quad Y_1^- \quad \dots \quad q_1^- \quad X^+ / \otimes q_i^+$$

Example



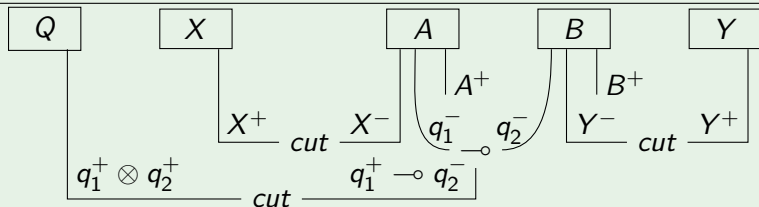
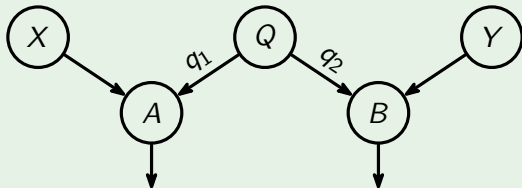
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Example



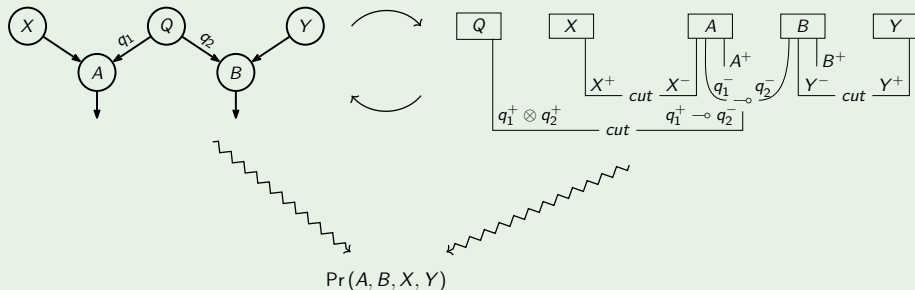
Exact Characterization by Proof-Nets

Theorem

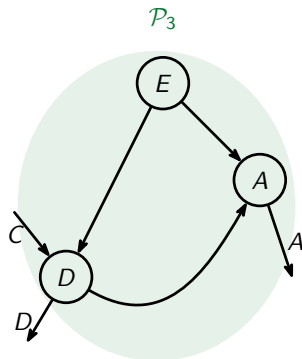
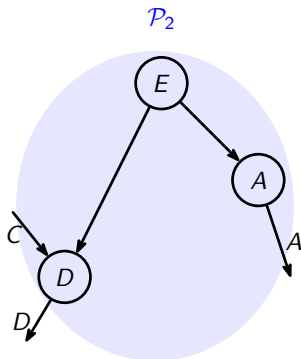
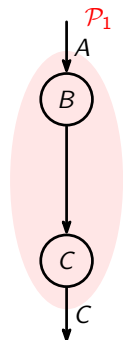
There is a **one-to-one correspondence** between Quantum Bayesian Networks and (a class of) Proof-Nets.

Furthermore, this correspondence **preserves the semantics** = the associated probability distribution.

Example

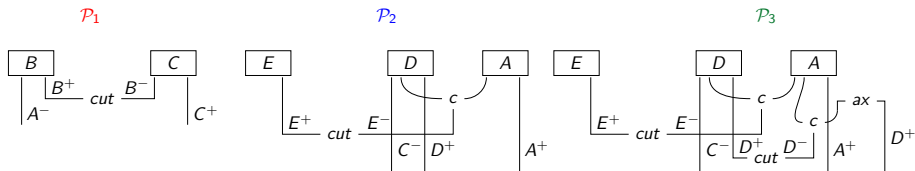


Modularity by Typing



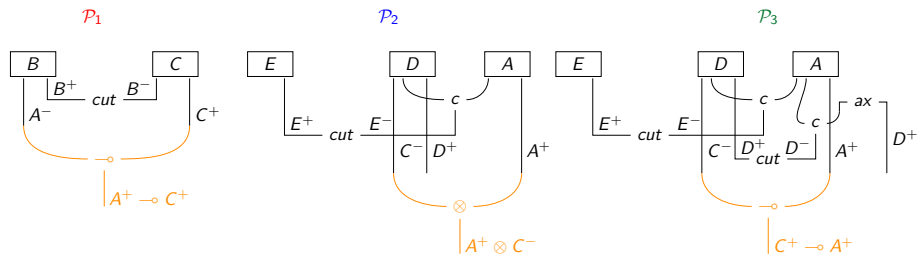
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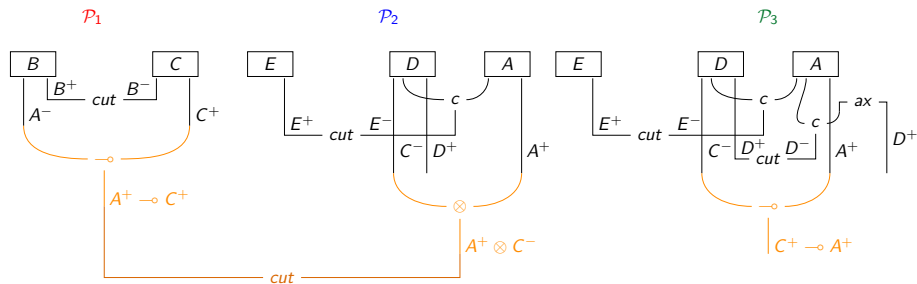
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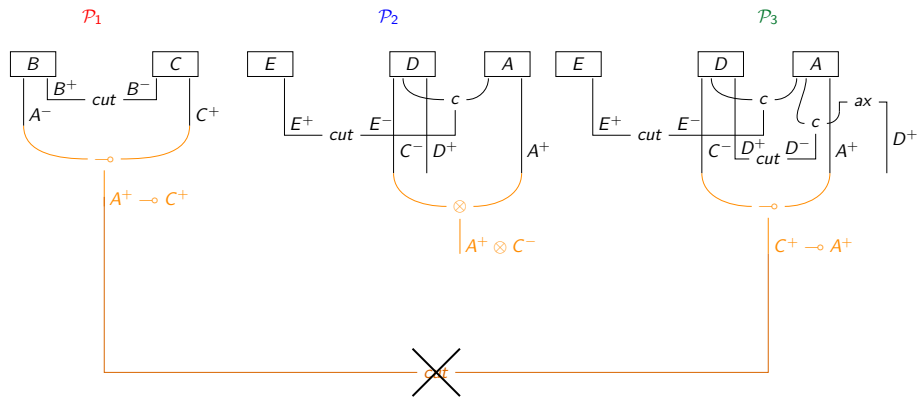
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Conclusion

Contributions

- **Compositionality** by modifying the semantic:
from quantum instruments to (more general) **quantum factors**
- **Conservative** extension: our QBNs are **exactly** the usual Bayesian Networks in the absence of quantum nodes
- **Modularity** by typing in proof-nets:
proof-theoretic approach adding an **interface**, parts with compatible interfaces are those giving a QBN

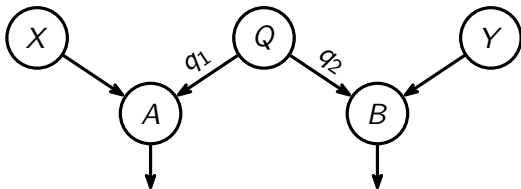
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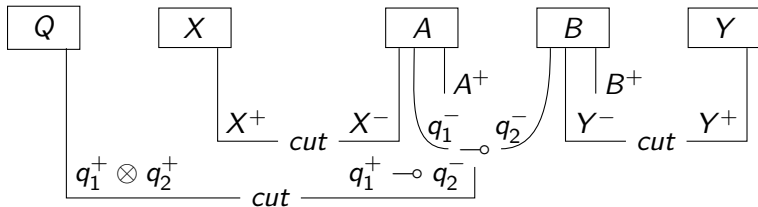
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Perspectives

- Using **terms** instead of proof-nets for typing
→ work started with Claudia Faggian and Gabriele Tedeschi
- Compositionality could be used to study **conditional independence (no-signaling)** between random variables or quantum causes



Thank you for
your attention!



References



Henson, Joe, Raymond Lal, and Matthew F Pusey (Nov. 2014).
“Theory-independent limits on correlations from generalized Bayesian networks”. In: *New Journal of Physics* 16.11. ISSN: 1367-2630. DOI: 10.1088/1367-2630/16/11/113043.